Financial Management and Valuation
Unit 4: Mini Cases

SBWL Finance

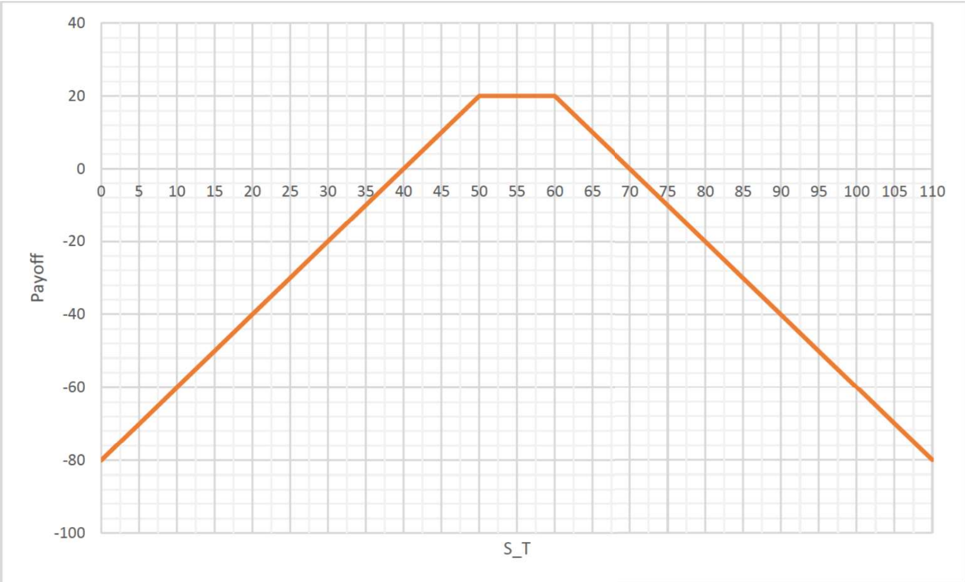
Winter Term 2025-26

Please upload your solution and indicate which Mini-Cases (MCs) you have solved before January 12, 2026 at 10pm. To do so, go to the assignment section on Canvas and choose "Unit 4 Indication and Submission of MCs". Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions file upload will be considered, your last complete attempt counts.**

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism.

The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Mini-Case 4.1:
Consider the following diagram which displays the payoff of a portfolio of European call and put options at maturity T:



a) Which positions in European calls and puts do you need to enter to achieve the payoff above? State the exact number of contracts, the type of position (long/short) and the corresponding strike prices.

b) Draw the payoff diagram at maturity T of each portfolio component in a) individually and show how you can derive the portfolio payoff step by step.

Mini-Case 4.2:
You are an Austrian entrepreneur who is expecting an inflow of 50.000 GBP in three months. In the past, you entered a short forward contract to sell GBP for EUR that will expire in three months with a contract size of 50.000 GBP to hedge your inflow of foreign currency. The original forward price was 1.18 EUR for 1 unit of GBP. The current exchange rate is 1.14 EUR/GBP. Assume a domestic and foreign linear risk-free interest rate of 2.5%p.a. and 4%p.a. respectively.

a) What is the current EUR/GBP forward price with a maturity in three months?

b) What is the market value of the outstanding forward contract?

Mini-Case 4.3:
You are interested in pricing both a European call to buy GBP with USD and a European put to sell GBP for USD.

a) Use the market data presented on the lecture slides and apply a one-period binomial model to calculate the price of a call and a put with $K = 1.30$ [USD] with maturity 3 months. For your calculations, compute the mid-values from the bid- and ask-quotes of the composite spot, the ATM implied volatility, US and UK deposit rates.

b) Show that put-call-parity holds.

Mini-Case 4.4:
Suppose an investor buys one European put with strike 0.8000 and sells a European put with strike 0.6000. The two puts are written on the same underlying asset and have the same maturity.

a) Draw the payoff of the trading strategy as a function of the underlying.
b) Implementing the trading strategy costs 1255 pips. For which value of the underlying asset does the strategy have a break even in the profits (i.e., the value of the underlying for which profits are zero)?
c) How does the break even of the profits change if the long put in the trading strategy has a strike of 0.9000 and the price of the trading strategy remains 1255 pips? Is there an arbitrage?

Mini-Case 4.5:
Cryptostar, a firm mining a particular cryptocurrency, can mine 4.7 units of the cryptocurrency in year 1 and 3.5 units in year 2. The cost for each unit of currency, due to electricity bills and equipments, is estimated to be 320 for each unit of currency in period one and 510 for each unit of currency in period two. The current forward prices are 460 for a one-year contract and 700 for a two-year contract. The annually compounded risk-free rates are 5% for one year and 7% for two years. The firm sells the cryptocurrencies at the end of each year.

a) What are the cash flows of the firm at time 1 and 2?

b) Find the tracking portfolio that replicates the cash flows in a).

c) Use the tracking portfolio to find the value of the firm.

Mini-Case 4.6:
A brewery is considering entering the soft drink market and can use existing machines for production. The company has no free capacity this period (t=0), but knows that it will have enough free capacity to produce $400,000$ soft drinks in each of t=1 and t=2. While the sale price is expected to remain stable at €2.50 per soft drink over the next 2 years, production costs are expected to vary. Currently the production costs would be €1.20 per

soft drink, in the coming two years they can increase by 50% or decrease by 10% from the previous year's cost.

A market portfolio in the soft drink industry over the next two years depends entirely on the production costs. If production costs increase, the value of the market portfolio decreases by 25%. If production costs decrease, the value of the market portfolio increases by 25%.

The risk-free return is 2.5%.

a) Value the opportunity to enter the soft drink market. Assume that the company can decide whether to enter or not in period t=0 and thereafter cannot change this decision.

b) Assume that the brewery has some flexibility. Specifically, if the production cost exceeds the selling price of a soft drink, the brewery can simply stop producing without additional costs. Furthermore, in t=2, the brewery has some additional free capacity and can therefore increase production from 400,000 to 600,000. What is the value of this additional flexibility in production?